Term 4 Checklist

Topic	Learning Outcomes	RAG	
		1	2
Y7 Electricity (REVISION)	Describe how charged particles interact		
	Describe what causes static electricity		
	State which particle moves in an electrical current		
	Compare electrical conductors and insulators		
	Use a simple model to describe the difference between current and potential difference (voltage)		
	State the units of current and potential difference		
	Explain why current is the same at all points in a circuit		
	Describe the difference between a series and parallel circuit (Set 1)		
	Calculate the current in a series and parallel circuit (Set 1)		
	State how to measure current and potential difference		
	Draw complete circuit diagrams using a rule, pencil and straight lines		
	Draw a range of common circuit symbols e.g. wire, voltmeter, bulb		
	Describe the difference between AC and DC and give an example of the uses of each		
Y7 Interdependence (REVISION)	Draw a food chain and web for a described feeding relationship		
	Identify producers, consumers and decomposers in a feeding relationship		
	Predict the effect of an environmental change on a food web		
	State the resources that animals and plants compete for		
	Explain why some animals are more successful competitors		
	Describe how plants can spread their seeds and why it's important		
	Describe the role of pollen and how plants reproduce		
	Explain how pollinators are useful to humans		
	Give the word equation for photosynthesis		
	Explain how plants are adapted for photosynthesis		
	Describe the roles of different plant structures e.g. leaves, roots, stems, flowers		
Y8 Human Body (REVISION)	Define diffusion and give examples of diffusion in the human body		
	State the 3 main nutrients and describe their uses in the body		
	Explain how temperature and concentration changes affect rate of diffusion		
	Name the major digestive organs and describe their functions		
	Describe the order food travels through the digestive system		
	Explain the importance of villi in the digestive system		
	Describe the function of the circulatory system		
	Describe the structure and function of the heart		
	Explain the different thicknesses of the left and right walls of the heart		
	Explain why a high fat diet can lead to an increase in blood pressure		
	Label the main parts of the lungs on a diagram		
	Describe and explain the adaptations shown by the alveoli		
	Explain how breathing works in terms of changes in pressure in the chest		
Periodic Table	Describe what an atom is and state their approximate size		
	Compare atoms and elements		

	Draw particle diagrams showing elements and compounds		
	Describe what a semi-metal is		
	Describe the general properties of metals and non-metals		
	Describe how the periodic table is arranged in groups and periods		
	Identify the group and period of an element on the periodic table		
	Define "subatomic particle"		
	Draw a labelled diagram of an atom including: nucleus, protons, neutrons, shells and electrons		
	State the charge of the three subatomic particles		
	Use the periodic table to calculate the number of protons, neutrons and electrons in an atom		
	State the names of groups 1, 2, 7 and 8 of the periodic table		
	Describe the properties of elements in group 7 (HT only)		
	Describe the trend in reactivity down group 7 (HT only)		
	Explain the trend in reactivity down group 7 in terms of electronic structure (HT only)		
	Explain why displacement reactions occur between metals and metal compounds		
	Describe the test and positive results for the gases: hydrogen, chlorine, carbon dioxide and oxygen		
	Describe how to collect gases using a submerged measuring cylinder		
	Describe how to calculate rate of production of gas per second		
	Describe a range of simple test tube reactions that can be used to identify an unknown substance		
	Explain the difference in properties between compounds and mixtures		
	Define density for solids in terms of mass and volume		
	Define density for liquids and gases in terms of particles		
	Explain why water sinks to the bottom of a beaker of oil		
	Explain why you would float easily in very salty water, but not in a swimming pool		
Staying Alive	Describe the role of stem cells in the body		
	Compare asexual and sexual reproduction		
	Describe the advantages and disadvantages of asexual reproduction		
	Describe the steps involved in plant reproduction (pollination, fertilisation, germination)		
	Label the reproductive parts of a plant		
	Describe how fertilisation happens in animals		
	Compare gestation periods of animals to their size		
	Describe the main changes that occur during puberty		
	Describe how puberty is controlled by hormones		
	State the two main sex hormones and where they are produced		
	Label the male reproductive organs		
	Label the female reproductive organs		
	Describe the function of the male reproductive system		
	Describe the function of the female reproductive system		
	Describe the steps involved in the menstrual cycle		

	Describe the impact of the menopause on female fertility		
	Describe the uses of contraception		
	Explain the importance of consent in a sexual relationship		
	Explain why condoms are not 100% effective at preventing pregnancy and STDs		
	Describe how oxygen and glucose reach a growing foetus		
	Explain the impact of smoking and vaping on foetal development		
	Describe what an STD is and give examples of common STDs		
	Describe the role of DNA in cells and where it is found		
	Describe the main steps in extracting DNA		
	Describe the two causes of variation in organisms		
	Give examples of variation caused by environmental and genetic factors		
	Describe artificial, sexual and natural selection		
	Explain how evolution by natural selection occurs		
	Give examples of adaptations in humans that have been naturally selected		